

Q1 - 24 June - Shift 1

Bag A contains 2 white, 1 black and 3 red balls and bag B contains 3 black, 2 red and n white balls. One bag is chosen at random and 2 balls drawn from it at random, are found to be 1 red and 1 black. If the probability that both balls come from Bag A is $\frac{6}{11}$, then n is equal to _____.

(A) 13

(B) 6

(C) 4

(D) 3

Space for your notes:

Q2 - 24 June - Shift 1

If a random variable X follows the Binomial distribution $B(33, p)$ such that $3P(X = 0) = P(X = 1)$, then the value of $\frac{P(X = 15)}{P(X = 18)} - \frac{P(X = 16)}{P(X = 17)}$ is equal to

(A) 1320

(B) 1088

(C) $\frac{120}{1331}$ (D) $\frac{1088}{1089}$ *Space for your notes:*

Q3 - 24 June - Shift 2

Questions

MathonGo

A random variable X has the following probability distribution :

X	0	1	2	3	4
P(X)	k	2k	4k	6k	8k

The value of $P(1 < X < 4 \mid X \leq 2)$ is equal to :

(A) $\frac{4}{7}$

(B) $\frac{2}{3}$

(C) $\frac{3}{7}$

(D) $\frac{4}{5}$

Space for your notes:

Q4 - 24 June - Shift 2

In an examination, there are 10 true-false type questions. Out of 10, a student can guess the answer of 4 questions correctly with probability $\frac{3}{4}$ and the remaining 6 questions correctly with probability $\frac{1}{4}$.

If the probability that the student guesses the answers of exactly 8 questions correctly out of 10 is $\frac{27k}{4^{10}}$, then k is equal to _____.

Space for your notes:

Q5 - 25 June - Shift 1

#MathBoleTohMathonGo

Let E_1 and E_2 be two events such that the conditional probabilities $P(E_1|E_2) = \frac{1}{2}$,

$P(E_2|E_1) = \frac{3}{4}$ and $P(E_1 \cap E_2) = \frac{1}{8}$. Then:

(A) $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$

(B) $P(E'_1 \cap E'_2) = P(E'_1) \cdot P(E'_2)$

(C) $P(E_1 \cap E'_2) = P(E_1) \cdot P(E_2)$

(D) $P(E'_1 \cap E_2) = P(E_1) \cdot P(E_2)$

Space for your notes:

Q6 - 25 June - Shift 2

A biased die is marked with numbers 2, 4, 8, 16, 32, 32 on its faces and the probability of getting a face with mark n is $\frac{1}{n}$. If the die is thrown thrice, then the probability, that the sum of the numbers obtained is 48, is

(A) $\frac{7}{2^{11}}$

(B) $\frac{7}{2^{12}}$

(C) $\frac{3}{2^{10}}$

(D) $\frac{13}{2^{12}}$

Space for your notes:

Q7 - 26 June - Shift 1

Questions

MathonGo

Let a biased coin be tossed 5 times. If the probability of getting 4 heads is equal to the probability of getting 5 heads, then the probability of getting atmost two heads is:

- (A) $\frac{275}{6^5}$ (B) $\frac{36}{5^4}$ (C) $\frac{181}{5^5}$ (D) $\frac{46}{6^4}$

*Space for your notes:***Q8 - 26 June - Shift 2**

If the probability that a randomly chosen 6-digit number formed by using digits 1 and 8 only is a multiple of 21 is p, then 96 p is equal to _____.

*Space for your notes:***Q9 - 27 June - Shift 1**

Five numbers x_1, x_2, x_3, x_4, x_5 are randomly selected from the numbers 1, 2, 3,....., 18 and are arranged in the increasing order ($x_1 < x_2 < x_3 < x_4 < x_5$). The probability that $x_2 = 7$ and $x_4 = 11$ is :

- (A) $\frac{1}{136}$ (B) $\frac{1}{72}$
(C) $\frac{1}{68}$ (D) $\frac{1}{34}$

*Space for your notes:***Q10 - 27 June - Shift 1**

#MathBoleTohMathonGo

Questions

MathonGo

Let X be a random variable having binomial distribution $B(7, p)$. If $P(X = 3) = 5P(X = 4)$, then the sum of the mean and the variance of X is :

Space for your notes:

(A) $\frac{105}{16}$

(B) $\frac{7}{16}$

(C) $\frac{77}{36}$

(D) $\frac{49}{16}$

Q11 - 27 June - Shift 2

If a point $A(x, y)$ lies in the region bounded by the y -axis, straight lines $2y + x = 6$ and $5x - 6y = 30$, then the probability that $y < 1$ is :

Space for your notes:

(A) $\frac{1}{6}$

(B) $\frac{5}{6}$

(C) $\frac{2}{3}$

(D) $\frac{6}{7}$

Q12 - 27 June - Shift 2

Let $S = \{E_1, E_2, \dots, E_8\}$ be a sample space of random

Space for your notes:

experiment such that $P(E_n) = \frac{n}{36}$ for every

$n = 1, 2, \dots, 8$. Then the number of elements in the

set $\left\{ A \subset S : P(A) \geq \frac{4}{5} \right\}$ is _____

Q13 - 28 June - Shift 1

The probability, that in a randomly selected 3-digit number at least two digits are odd, is

Space for your notes:

(A) $\frac{19}{36}$

(B) $\frac{15}{36}$

(C) $\frac{13}{36}$

(D) $\frac{23}{36}$

Answer Key

Q1 (C)

Q2 (A)

Q3 (A)

Q4 (479)

Q5 (C)

Q6 (D)

Q7 (D)

Q8 (33)

Q9 (C)

Q10 (C)

Q11 (B)

Q12 (19)

Q13 (A)

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Q1 (C) E_1 = denotes selection for 1st bag E_2 = denotes selection for 2nd bag

$$P(E_1) = \frac{1}{2}, P(E_2) = \frac{1}{2}$$

A = selected balls are 1 red & 1 black

$$P\left(\frac{A}{E_1}\right) = \frac{{}^3C_1 \times {}^1C_1}{{}^6C_2} = \frac{1}{5}$$

$$P\left(\frac{A}{E_1}\right) = \frac{{}^3C_1 \times {}^2C_1}{(n+5)C_2} = \frac{12}{(n+5)(n+4)}$$

$$P\left(\frac{E_1}{A}\right) = \frac{P(E_1) \times P\left(\frac{A}{E_1}\right)}{P(E_1) \times P\left(\frac{A}{E_1}\right) + P(E_2) \times P\left(\frac{A}{E_2}\right)}$$

$$= \frac{\frac{1}{10}}{\frac{1}{10} + \frac{6}{(n+5)(n+4)}} = \frac{6}{11}$$

$$\Rightarrow n = 4$$

Q2 (A) $n = 33$, let probability of success is p and $q = 1 - p$

$$3p(x=0) = p(x=1)$$

$$3 \cdot {}^{33}C_0 (q)^{33} = {}^{33}C_1 p q^{32}$$

$$p = \frac{1}{12}, q = \frac{11}{12}, \frac{q}{p} = 11$$

$$\frac{p(x=15)}{p(x=18)} = \frac{p(x=16)}{p(x=17)}$$

$$\frac{{}^{33}C_{15} p^{15} q^{18}}{{}^{33}C_{18} p^{18} q^{15}} = \frac{{}^{33}C_{16} p^{16} q^{17}}{{}^{33}C_{17} p^{17} q^{16}} = \left(\frac{q}{p}\right)^3 - \left(\frac{q}{p}\right)$$

$$= (11)^3 - 11$$

$$= 1320$$

#MathBoleTohMathonGo

Q3 (A)

$$P\left(\frac{1 < x < 4}{x \leq 2}\right) = \frac{P(1 < x < 4 \cap x \leq 2)}{P(x \leq 2)}$$

$$= \frac{P(1 < x \leq 2)}{P(x \leq 2)} = \frac{P(x = 2)}{P(x \leq 2)}$$

$$= \frac{4k}{k + 2k + 4k} = \frac{4}{7}$$

Q4 (479)

$$A = \{1, 2, 3, 4\} : P(A) = \frac{3}{4} \rightarrow \text{Correct}$$

$$B = \{5, 6, 7, 8, 9, 10\} ; P(B) = \frac{1}{4} \text{ Correct}$$

8 Correct Ans.:

$$(4, 4): {}^4C_4 \left(\frac{3}{4}\right)^4 \cdot {}^6C_4 \cdot \left(\frac{1}{4}\right)^4 \cdot \left(\frac{3}{4}\right)^2$$

$$(3, 5): {}^4C_3 \left(\frac{3}{4}\right)^3 \cdot \left(\frac{1}{4}\right)^1 \cdot {}^6C_5 \left(\frac{1}{4}\right)^5 \cdot \left(\frac{3}{4}\right)$$

$$(2, 6): {}^4C_2 \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^2 \cdot {}^6C_6 \left(\frac{1}{4}\right)^6$$

$$\text{Total} = \frac{1}{4^{10}} [3^4 \times 15 \times 3^2 + 4 \times 3^3 \times 6 \times 3 + 6 \times 3^2]$$

$$= \frac{27}{4^{10}} [2.7 \times 15 + 72 + 2]$$

$$\Rightarrow K = 479$$

Q5 (C)

$$(A) \quad P(E_1) \cdot P(E_2) = \frac{1}{6} \cdot \frac{1}{4} = \frac{1}{24} \neq P(E_1 \cap E_2)$$

$$\begin{aligned} (B) \quad P(E'_1 \cap E'_2) &= 1 - P(E_1 \cup E_2) \\ &= 1 - (P(E_1) + P(E_2) - P(E_1 \cap E_2)) \\ &= 1 - \left(\frac{1}{6} + \frac{1}{4} - \frac{1}{8} \right) = \frac{17}{24} \end{aligned}$$

$$P(E'_1)P(E_2) = \frac{5}{6} \times \frac{1}{4} = \frac{5}{24}$$

$$(C) \quad P(E_1 \cap E'_2) = P(E_1) - P(E_1 \cap E_2) = \frac{1}{6} - \frac{1}{8} = \frac{1}{24}$$

$$(D) \quad P(E'_1 \cap E_2) = P(E_2) - P(E_1 \cap E_2) = \frac{1}{4} - \frac{1}{8} = \frac{1}{8}$$

Q6 (D)

$$P(n) = \frac{1}{n}$$

$$P(2) = \frac{1}{2} \quad P(8) = \frac{1}{8}$$

$$P(4) = \frac{1}{4} \quad P(16) = \frac{1}{16}$$

$$P(32) = \frac{2}{32}$$

Possible cases

16, 16, 16 and 32, 8, 8

$$\text{Probability} = \frac{1}{16^3} + \frac{2}{32} \times \frac{1}{8} \times \frac{1}{8} \times 3 = \frac{13}{16^3}$$

Q7 (D)

Hints and Solutions

MathonGo

$$P(H) = x, P(T) = 1 - x$$

$$P(4H, 1T) = P(5H)$$

$${}^5C_1(x)^4(1-x)^1 = {}^5C_5x^5$$

$$5(1-x) = x$$

$$6x = 5 \Rightarrow x = \frac{5}{6}$$

$$P(\text{atmost 2H})$$

$$= P(0H, 5T) + P(1H, 4T) + P(2H, 3T)$$

$$= {}^5C_0\left(\frac{1}{6}\right)^5 + {}^5C_1\frac{5}{6}\left(\frac{1}{6}\right)^4 + {}^5C_2\left(\frac{5}{6}\right)^3\left(\frac{1}{6}\right)^2$$

$$= \frac{1}{6^5}(1 + 25 + 250) = \frac{276}{6^5}$$

$$= \frac{46}{6^4}$$

Q8 (33)

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$$2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$$

Divisible by 21 when divided by 3.

Case – I : All 1 $\rightarrow$ (1)

Case – II : All 8 $\rightarrow$ (1)

Case – III : 3 ones & 3 eights

$$\frac{6!}{3! \times 3!} = 20$$

Required probability $\therefore p = \frac{22}{64}$

$$96p = 96 \times \frac{22}{64} = 33$$

Q9 (C)

No. of ways to select and arrange x_1, x_2, x_3, x_4, x_5
from 1, 2, 3, 18

$$n(s) = {}^{18}C_5$$

$$x_1 \quad (x_2) \quad x_3 \quad (x_4) \quad x_5$$

$$n(E) = {}^6C_1 \times {}^3C_1 \times {}^7C_1$$

$$P(E) = \frac{6 \times 3 \times 7}{{}^{18}C_5}$$

$$\frac{1}{17 \times 4} = \frac{1}{68}$$

Q10 (C)

B (7, p)

$$n = 7 \quad p = p$$

given

$$P(x = 3) = 5P(x = 4)$$

$${}^7C_3 \times p^3(1-p)^4 = 5 \cdot {}^7C_4 p^4(1-p)^3$$

$$\frac{{}^7C_3}{5 \times {}^7C_4} = \frac{p}{1-p}$$

$$1 - p = 5p$$

$$6p = 1$$

$$p = \frac{1}{6} \Rightarrow q = \frac{5}{6}$$

$$n = 7$$

$$\text{Mean} = np = 7 \times \frac{1}{6} = \frac{7}{6}$$

$$\text{Var} = npq = 7 \times \frac{1}{6} \times \frac{5}{6} = \frac{35}{36}$$

Sum

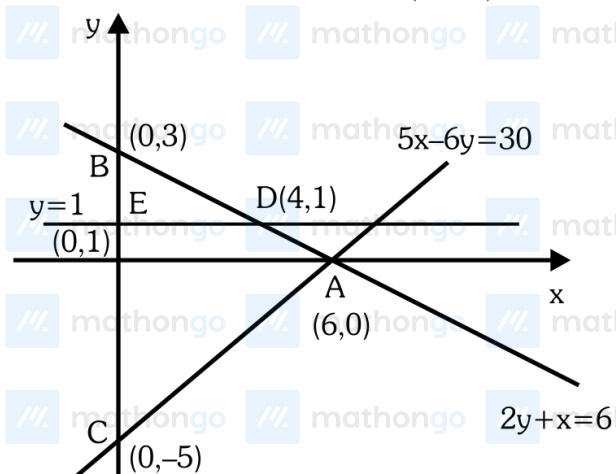
$$= \frac{7}{6} + \frac{35}{36}$$

$$= \frac{42 + 35}{36}$$

$$= \frac{77}{36}$$

Q11 (B)

$$\text{Required probability} = \frac{\text{ar}(\text{ADEC})}{\text{ar}(\text{ABC})}$$



$$= 1 - \frac{\text{ar}(\text{BDE})}{\text{ar}(\text{ABC})}$$

$$= 1 - \frac{\frac{1}{2} \times 2 \times 4}{\frac{1}{2} \times 8 \times 6} = 1 - \frac{1}{6} = \frac{5}{6}$$

Q12 (19)

Hints and Solutions

MathonGo

$$P(A') < \frac{1}{5} = \frac{36}{180}$$

5 times the sum of missing number should be less than 36.

If 1 digit is missing = 7

If 2 digit is missing = 9

If 3 digit is missing = 2

If 0 digit is missing = 1

Alternate

A is subset of S hence

A can have elements:

type 1 : { }

type 2: {E₁}, {E₂}, {E₈}

type 3: {E₁, E₂}, {E₁, E₃}..... {E₁, E₈}

⋮

type 6: {E₁, E₂, E₅}, {E₄, E₅, E₆, E₇, E₈}

type 7: {E₁, E₂, E₆}, {E₃, E₄, E₈}

type 8: {E₁, E₂, E₇} {E₂, E₃, E₈}

type 9: {E₁, E₂, E₈}

$$\text{As } P(A) \geq \frac{4}{5};$$

Note : Type 1 to Type 4 elements can not be in set

A as maximum probability of type 4 elements.

$$\{E_5, E_6, E_7, E_8\} \text{ is } \frac{5}{36} + \frac{6}{36} + \frac{7}{36} + \frac{8}{36} = \frac{13}{18} < \frac{4}{5}$$

Now for Type 5 acceptable elements let's call probability as P₅

$$P_5 = \frac{n_1 + n_2 + n_3 + n_4 + n_5}{36} \leq \frac{4}{5}$$

$$\Rightarrow n_1 + n_2 + n_3 + n_4 + n_5 \geq 28.8$$

Hence, 2 possible ways {E₅, E₆, E₇, E₈, E₃ or E₄}

$$P_6 = n_1 + n_2 + n_3 + n_4 + n_5 + n_6 \geq 28.8$$

$\Rightarrow$ 9 possible ways

$$P_7 \Rightarrow n_1 + n_2 + \dots + n_7 \geq 28.8$$

$\Rightarrow$ 7 possible ways

$$P_8 \Rightarrow n_1 + n_2 + \dots + n_8 \geq 28.8$$

$\Rightarrow$ 1 possible way

Total = 19

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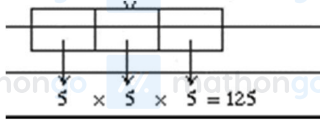
Q13 (A)

Atleast two digits are odd

= exactly two digits are odd + exactly there 3 digits

are odd

For exactly three digits are odd



For exactly two digits odd :

If 0 is used then : $2 \times 5 \times 5 = 50$

If 0 is not used then : ${}^3C_1 \times 4 \times 5 \times 5 = 300$

Required Probability = $\frac{475}{900} = \frac{19}{36}$